FORENSIC AUDIT EXECUTIVE REPORT

Dataset Evaluated: **clean_synthetic_fraud_dataset.csv** | Status: Complete

Phase 3 Baseline Score	Phase 4 Forensic Score	P3 Flagged	P4 Flagged	Newly Surfaced
31.26 / 100	18.68 / 100	6476	2784	1404

Forensic Test Breakdown

Test Executed	Calculated Risk Score	Records Flagged
Correlation Analysis	0.00	0
Outlier Detection	27.84	2784
Benford's Law	28.21	0

Executive Findings & Recommendations

Executive Risk Assessment

The system exhibits **High Risk** due to a significant number of flagged transactions (6,476 in Phase 3 and 2,784 in Phase 4) despite low baseline and forensic scores (31.26/100 and 18.68/100, respectively). The overlap of 1,380 flagged records suggests persistent high-risk patterns, while the 1,404 newly surfaced cases indicate evolving fraud tactics. The presence of 14 high-risk records with confirmed fraud labels (Fraud_Label = 1) further validates the elevated risk profile.

Key Anomalies Identified

1. **Outlier Detection Dominance**: The forensic test with the highest flagged count (2,784) is Outlier Detection, which may indicate systemic anomalies in transaction patterns or data inconsistencies.
2. **High Risk Scores with Fraud Labels**: Sample records (e.g., TXN_40623 with Risk_Score = 0.9977 and Fraud_Label = 1) show the model effectively identifying fraudulent activity, but the low forensic score (18.68/100) suggests potential calibration issues.
3. **IP Address Discrepancies**: Records with IP_Address_Flag = 1 (e.g., TXN_44118) are flagged but not labeled as fraud, raising concerns about false positives or incomplete data validation.
4. **Transaction Distance Anomalies**: High Transaction_Distance values (e.g., 3,224.94 km for TXN_6678) may indicate location spoofing or device misuse.
5. **High Transaction Volume**: Records with elevated Daily_Transaction_Count (e.g., 10 transactions for TXN_40623) and Failed_Transaction_Count_7d (e.g., 4 failures for TXN_6678) align with known fraud indicators.

Actionable Recommendations for Audit Team

1. **Validate Outlier Detection Logic**: Investigate why Outlier Detection flags 2,784 records. Verify if these anomalies stem from legitimate data variability or undetected fraud patterns.
2. **Re-evaluate Scoring Model Calibration**: The low Phase 4 forensic score (18.68/100) contrasts with high-risk labels. Adjust scoring thresholds to better align with actual fraud detection efficacy.
3. **Audit IP Address Flagging**: Review discrepancies in IP_Address_Flag (e.g., TXN_44118) to determine if false positives are due to incomplete geolocation data or misclassification.
4. **Prioritize High-Risk Records**: Focus audit efforts on records with Fraud_Label = 1 (e.g., TXN_6678, TXN_40623) to validate fraud patterns and refine detection rules.
5. **Monitor Transaction Distance Metrics**: Analyze Transaction_Distance anomalies to assess whether location spoofing or device misuse is prevalent.
6. **Investigate High Transaction Volumes**: Scrutinize records with elevated Daily_Transaction_Count and Failed_Transaction_Count_7d to identify potential fraud attempts or system errors.
7. **Track Newly Surfaces Cases**: The 1,404 newly surfaced records may represent emerging fraud tactics. Conduct a root-cause analysis to update detection strategies.